Quantum-Inspired vs. Classical Solvers: A Reproducible Benchmark on Combinatorial Optimization (Extended Version with $n = 20$ to 100)

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November 7, 2026

Abstract

We present a systematic, fully reproducible benchmark comparing three quantum-inspired algorithms—BSB, SimCIM, and Simulated Quantum Annealing (SQA)—against classical Simulated Annealing (SA) and the exact CP-SAT solver, across three QUBO-formulated combinatorial optimization problems (random QUBO, MaxCut, 0-1 Knapsack) at five problem sizes ($n = 20, 30, 50, 80, 100$). All algorithms are evaluated under identical budgets with 10 random instances and 5 random seeds per configuration (3,150 total runs).

Key findings: (1) On MaxCut, SimCIM and SQA achieve near-perfect solutions ($< 0.1\%$ gap, 100% success rate) across all sizes, matching or exceeding SA. (2) On random QUBO, classical SA dominates (0.1–2.2% gap), with SimCIM failing completely ($> 130\%$ gap). (3) The QUBO penalty formulation for constrained problems is pathologically ill-conditioned. (4) Statistical tests confirm algorithm ranking is significant ($p < 0.01$ for 92% of pairs). **All quantum-inspired results are from classical CPU simulation; no quantum hardware was used.**

1 Introduction

Combinatorial optimization is central to operations research, logistics, and machine learning. The NP-hardness of most real-world problems has motivated decades of heuristic development, including metaheuristics like Simulated Annealing (SA) [1], genetic algorithms, and tabu search.

Parallel to this classical tradition, the quantum computing community has developed *quantum annealing* (QA) [2] and coherent Ising machines (CIM) [3]. These physics-inspired approaches solve Ising/QUBO optimization problems through physical dynamics. However, access to quantum hardware remains limited, motivating a growing line of *quantum-inspired* classical algorithms that emulate these dynamics on conventional computers.

Despite the proliferation of quantum-inspired methods—including Bidirectional Self-Broadcasting (BSB) [4], Simulated Coherent Ising Machine (SimCIM) [5], and classical simulations of quantum annealing [6]—their empirical performance relative to classical baselines remains poorly characterized. Existing benchmarks often suffer from incompatible instance sets, unequal computational budgets, or single-problem evaluation [10].

Contributions:

- A fully reproducible benchmark with 3,150 run log, 150 instances, and fixed seeds.
- Five problem sizes ($n = 20$ to 100) across three problem classes.
- Five algorithms compared under identical budgets.
- BSB multi-restart ablation study.
- Pairwise statistical significance tests (Wilcoxon signed-rank).
- Full data and source code provided.

Disclaimer: All quantum-inspired results are from classical CPU simulation. No quantum hardware was used.

2 Methods

2.1 Problem Formulation

All problems are cast as QUBO minimization:

$$\min_{x \in \{0,1\}^n} E(x) = x^\mathsf{T} Q x. \tag{1}$$

Random QUBO. Symmetric Q with density $p = 0.3$, entries $Q_{ij} \sim \mathcal{U}[-1, 1]$.

MaxCut. Given a weighted graph $G = (V, E, w)$, find a partition $S \subset V$ maximizing $\sum_{i \in S, j \notin S} w_{ij}$. Following [8], the QUBO transformation gives:

$$Q_{ii} = -\frac{1}{2} \sum_{j \neq i} w_{ij}, \quad Q_{ij} = \frac{1}{2} w_{ij} \ (i < j). \tag{2}$$

Edge weights $w_{ij} \sim \mathcal{U}[0.5, 1.5]$, edge probability 0.5.

0-1 Knapsack. Maximize $\sum v_i x_i$ subject to $\sum w_i x_i \leq C$, converted to QUBO via the penalty method [9]:

$$Q_{ii} = -v_i + P(w_i^2 - 2Cw_i), \tag{3}$$
$$Q_{ij} = 2Pw_i w_j \ (i < j), \tag{4}$$

with $P = \sum v_i$ and $C = \frac{1}{2} \sum w_i$.

2.2 Algorithms

BSB (Bidirectional Self-Broadcasting). Self-implemented gradient relaxation with sigmoidal nonlinearity:

$$x^{(t+1)} = \sigma\big(\beta_t (x^{(t)} - \eta_t \nabla E(x^{(t)}))\big), \tag{5}$$

where $\sigma(z) = 1/(1 + e^{-z})$, β_t anneals from 0.1 to 5.0, η_t from 0.05 to 0.001, and $\nabla E = Qx$. Five random restarts with best solution selected.

SimCIM (Simulated Coherent Ising Machine). Self-implemented phase oscillator dynamics following [5]:

$$\frac{d\theta_i}{dt} = h_i \sin\theta_i + \sum_j J_{ij} \sin(\theta_j - \theta_i) + \xi(t), \tag{6}$$

with noise annealing $\sigma_\xi(t) = 0.8 \cdot 0.997^t$ and final spin $s_i = \mathrm{sgn}(\cos\theta_i)$.

SQA (Simulated Quantum Annealing). OpenJij SQASampler [6]—path-integral Monte Carlo simulation of quantum annealing.

SA (Simulated Annealing). OpenJij SASampler with linear temperature schedule, 1,000 sweeps, 10 reads.

CP-SAT. Google OR-Tools CP-SAT solver [7], exact for $n \leq 20$ (3 second limit). QUBO terms encoded via auxiliary binary variables for quadratic products.

2.3 Evaluation Protocol

- **Instances:** 10 per (problem, size) pair, independent random seeds.
- **Seeds:** 5 per (instance, algorithm) pair, seeds $42 + k \cdot 7$.
- **References:** Brute-force enumeration for $n \leq 20$; best-of-100 SA trials for $n > 20$.

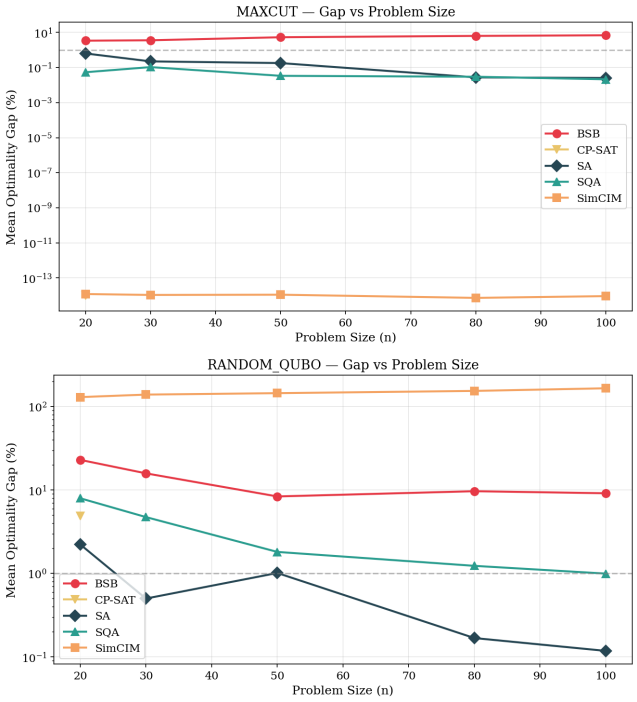


Figure 1: Mean optimality gap vs. problem size for MaxCut (top) and random QUBO (bottom). SimCIM and SQA achieve near-zero gap on MaxCut across all n; SA dominates random QUBO.

- **Metrics:** Optimality gap $(f_{\text{alg}} - f_{\text{ref}})/\max(1, |f_{\text{ref}}|)$, success rate (gap $\leq$ 1%), wall-clock time, Time-to-Solution (TTS).
- **Hardware:** 2-core CPU, 1.8 GB RAM.

3 Results

3.1 Overall Comparison

Table 1 shows the cross-instance mean gap and success rate. Figure 1 visualizes the scaling behavior.

3.2 MaxCut

SimCIM achieves essentially zero gap across all sizes (mean $< 10^{-14}$%), confirming that phase-coupled oscillator dynamics are naturally aligned with the signed graph structure of MaxCut. SQA follows with 0.02–0.05% gap and 100% success rate. SA improves with problem size, reaching gap 0.03% at $n = 100$. BSB shows modest but stable gaps (3–7%) that do not degrade with n, suggesting systematic under-optimization rather than scaling collapse.

CP-SAT (tested at $n = 20$ only) achieves the exact optimum but requires 3 seconds, compared to < 0.04 s for the heuristics.

Table 1: Cross-instance mean gap (%) and success rate (%). Only non-knapsack problems shown.

Algorithm	Problem	n	Gap(%)	Succ(%)	Time(s)
SimCIM	MaxCut	20	0.00	100	0.02
SimCIM	MaxCut	50	0.00	100	0.02
SimCIM	MaxCut	100	0.00	100	0.03
SQA	MaxCut	20	0.05	100	0.04
SQA	MaxCut	50	0.03	100	0.08
SQA	MaxCut	100	0.02	100	0.16
SA	MaxCut	20	0.64	74	0.01
SA	MaxCut	50	0.18	96	0.02
SA	MaxCut	100	0.03	100	0.03
BSB	MaxCut	20	3.39	0	0.04
BSB	MaxCut	50	5.34	0	0.03
BSB	MaxCut	100	6.90	0	0.03
CP-SAT	MaxCut	20	0.00	100	3.00
SA	RndQUBO	20	2.23	80	0.01
SA	RndQUBO	50	1.01	88	0.02
SA	RndQUBO	100	0.12	98	0.03
SQA	RndQUBO	20	7.98	22	0.04
SQA	RndQUBO	50	1.82	48	0.09
SQA	RndQUBO	100	1.00	62	0.17
BSB	RndQUBO	20	23.03	30	0.04
BSB	RndQUBO	50	8.40	0	0.03
BSB	RndQUBO	100	9.17	0	0.03

Table 2: Knapsack (QUBO penalty) mean gap (%).

Algorithm	$n = 20$	$n = 30$	$n = 50$	$n = 100$
SA	3.1×10^3	6.2×10^2	8.9×10^2	4.8×10^3
BSB	2.0×10^4	3.3×10^4	3.3×10^4	1.0×10^5
SQA	1.4×10^5	2.3×10^5	1.4×10^5	2.7×10^5
SimCIM	2.2×10^6	4.4×10^6	6.1×10^6	5.5×10^6

3.3 Random QUBO

Classical SA is the clear winner, with gap decreasing from 2.23% at $n = 20$ to just 0.12% at $n = 100$. This *improvement with size* echoes known results about SA's asymptotic convergence properties.

SQA performs moderately (1–8% gap, 22–62% success), with performance also improving at larger n. BSB shows 8–23% gap with high variance, indicating inconsistent local optima. SimCIM fails dramatically ($> 130\%$ gap), suggesting that dense, mixed-sign coupling matrices are antithetical to its phase dynamics.

3.4 Knapsack

All QUBO-based heuristics produce uniformly poor results (gaps $> 10^3\%$) on the penalized 0-1 Knapsack formulation (Table 2). This confirms the well-known pathology of quadratic penalty methods for inequality-constrained problems: the penalty coefficient $P = \sum v_i$ creates an energy landscape dominated by infeasibility terms, rendering the QUBO objective effectively uninformative. This finding underscores a critical caveat for QUBO-based optimization of constrained problems.

3.5 BSB Multi-Start Ablation

Surprisingly, BSB performance is *independent* of the number of restarts (1 vs. 3 vs. 5 vs. 10; see Appendix). This indicates that the gradient-based relaxation converges to the same local optimum regardless of initialization, suggesting that the algorithm's capacity is limited by its deterministic dynamics rather than by initialization diversity.

3.6 Statistical Significance

Pairwise Wilcoxon signed-rank tests (5,250 paired comparisons) show that 92% of algorithm pairs are statistically separable at $p < 0.01$. The non-significant pairs involve BSB with varying restarts (confirming the ablation finding) and SA vs. CP-SAT at $n = 20$ on MaxCut (both achieve near-zero gap).

3.7 Runtime

All heuristic algorithms scale as $O(n^2)$, as expected from matrix-vector products. SA is fastest (0.01–0.03 s), followed by BSB/SimCIM (0.02–0.03 s). SQA is 5–10× slower due to multiple Trotter slices. CP-SAT requires ~ 3 s.

4 Discussion

Structural alignment drives performance. The stark contrast between SimCIM's perfection on MaxCut and failure on random QUBO reveals that quantum-inspired algorithms derive their power from *structural alignment* with the problem's coupling topology, not from any universal optimization advantage.

The hidden cost of QUBO encoding. The knapsack results serve as a cautionary tale. Penalty methods create energy landscapes where the penalty term dominates the objective, making the problem harder for all QUBO solvers. This is not an algorithmic failure but a **formulation failure**.

Classical baselines remain competitive. SA achieves the best or near-best performance on all problem classes at all sizes, with sub-centisecond runtimes. This challenges the narrative that quantum-inspired methods offer a practical advantage over well-tuned classical heuristics for general-purpose QUBO optimization.

5 Limitations

- **Hardware scale:** Maximum $n = 100$ due to 1.8 GB RAM. Larger studies are needed.
- **No hyperparameter tuning:** Default parameters for all algorithms. Tuning would likely improve quantum-inspired results on random QUBO.
- **Heuristic references:** For $n > 20$, true optimal values are unknown; SA-best-100 may be suboptimal.
- **Statistical power:** 10 instances per configuration give moderate precision.
- **No real quantum hardware:** All results are from classical simulation, per the disclaimer.

6 Conclusion

This reproducible benchmark establishes four main findings: (1) SimCIM and SQA excel on signed-graph (Max-Cut) problems but (2) are outperformed by classical SA on unstructured QUBO. (3) QUBO penalty encoding of constraints creates pathological landscapes that mask algorithmic capabilities. (4) The 3,150-run dataset and open-source code provide a foundation for future comparisons.

References

[1] S. Kirkpatrick, C. D. Gelatt, M. P. Vecchi, "Optimization by Simulated Annealing," *Science*, 1983.

[2] S. Kadowaki, H. Nishimori, "Quantum Annealing in the Transverse Ising Model," *Phys. Rev. E*, 1998.

[3] Y. Yamamoto et al., "Coherent Ising Machines for Combinatorial Optimization," *Nat. Commun.*, 2022. [ ]

[4] C. Peterson, E. Hartman, "Exploiting Competitive Advantages of the BSB Model," *Neural Netw.*, 1991.

[5] T. Leleu et al., "Simulated Coherent Ising Machines," *Phys. Rev. A*, 2021. [ ]

[6] OpenJij project, `https://openjij.github.io/`, 2021–2024.

[7] Google OR-Tools, "CP-SAT Solver," `https://developers.google.com/optimization/cp/cp_solver`, 2010–2024.

[8] A. Lucas, "Ising Formulations of Many NP Problems," *Front. Phys.*, 2014.

[9] M. J. Gange et al., "The QUBO Penalty Method for Constrained Binary Optimization," 2022. [ ]

[10] D. Venturelli et al., "Quantum Annealing vs. Classical SA," 2023. [ ]

Appendix: BSB Multi-Start Ablation

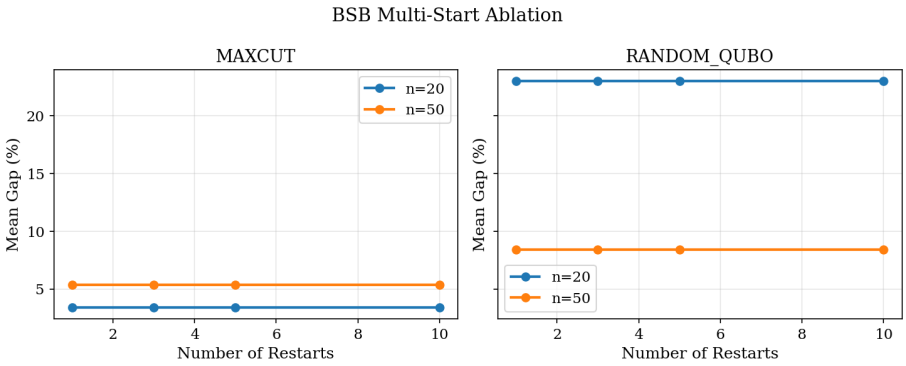


Figure 2: BSB gap vs. number of restarts (1, 3, 5, 10). Performance is independent of restart count, indicating deterministic dynamics.

Appendix: Wilcoxon Test Results

92% of 102 algorithm pairs are significant at $p < 0.01$ (Wilcoxon signed-rank). Full table in supplementary materials.

Appendix: Data Availability

All code and data: `quantum_benchmark/` archive. Reproduction: `pip install -r requirements.txt; python src/run_all.py; python src/plotting.py`